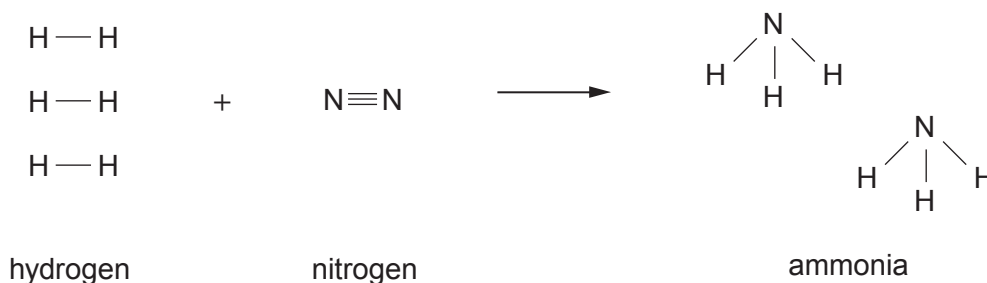


8. Ammonia is manufactured from hydrogen and nitrogen in the Haber process.

- (a) The equation shows the bonds which are broken and the bonds which are formed in the production of ammonia.

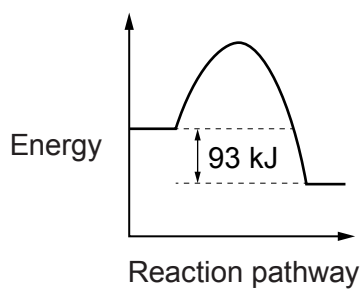


The bond energy of a H—H bond is 436 kJ. The energy needed to break **all** the bonds in the **reactants** is 2253 kJ.

- (i) Calculate the energy needed to break a N≡N bond. [2]

Energy = kJ

- (ii) The diagram below is the energy profile for the reaction.



Calculate the energy released when **all** the bonds in the ammonia molecules are formed. [1]

Energy = kJ



- (b) The table below shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)			
	200	300	400	500
	Percentage yield of ammonia			
100	81.7	52.5	25.2	10.6
200	89.0	66.7		18.3
400	94.6	79.7	55.4	31.9
1000	98.3	92.6	79.8	57.5

- (i) Describe the relationship between the percentage yield of ammonia and the temperature.

[1]

- (ii) **Circle** the most likely percentage yield of ammonia if the process were carried out at 200 atm and 400 °C.

[1]

20 %

30 %

40 %

50 %



- (c) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

Compound	Flame test	Add NaOH(aq)	Add HCl(aq)	Add BaCl ₂ (aq)	Add AgNO ₃ (aq)
A	no colour	pungent gas given off turns damp red litmus paper blue	no reaction	no reaction	white precipitate forms
B	green flame	blue precipitate forms	no reaction	white precipitate forms	no reaction
C	lilac flame	no reaction	gas given off turns limewater milky	no reaction	no reaction

Use the information provided to identify compounds **A**, **B** and **C**.

[3]

A

B

C

END OF PAPER

